

Independent Regulation of the Extracellular Cyclic AMP Phosphodiesterase-Inhibitor System and Membrane Differentiation by Exogenous Cyclic AMP in *Dictyostelium discoideum*

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Received December 27, 1977; accepted in revised form April 17, 1978

When amoebae of *Dictyostelium discoideum*, suspended in buffer, were treated with 100 nM pulses of cAMP, the extracellular cAMP phosphodiesterase (ePD) activity increased dramatically and the synthesis of the phosphodiesterase inhibitor (PDI) was repressed. In addition, the time of appearance on the cell surface of contact sites A, membrane-bound cAMP phosphodiesterase, and cAMP binding sites was accelerated by 3-4 hr and the concentration of intracellular cAMP increased ≈ 20 -fold. When the concentration of the cAMP pulse was reduced to 1 nM, the effect of the pulses on membrane differentiation and on the cAMP pool was virtually the same, while the effect on the ePD-PDI system was reduced. When cAMP was added to the suspension continuously, the nucleotide had no effect on membrane differentiation and failed to stimulate the intracellular cAMP pool, however, the ePD-PDI system was regulated normally. When the developmental mutant, HC112, was treated with cAMP pulses, membrane differentiation and the level of the cAMP pool were unaffected, while the ePD-PDI system responded to the exogenous cAMP. In another mutant, HC53, membrane differentiation was stimulated by cAMP pulses and this response was accompanied by a sharp increase in the concentration of the cAMP pool. These results suggest that the ePD-PDI system and membrane differentiation are regulated independently by exogenous cAMP and that regulation of the ePD-PDI system does not require activation of the adenyl cyclase.

INTRODUCTION

Upon starvation, amoebae of the cellular slime mold *Dictyostelium discoideum* enter the developmental phase of their life cycle. After a preaggregation period of 6-8 hr at 22°C, the cells stream into multicellular aggregates, proceed through various morphogenetic stages, and eventually form a fruiting body structure composed of highly differentiated stalk cells and spores (see Loomis, 1975). Both cell aggregation and cell differentiation have been shown to be regulated, at least in part, by the nucleotide adenosine 3'-5'-cyclic monophosphate (cAMP)¹ (Konijn *et al.*, 1968; Bonner, 1970;

Darmon *et al.*, 1975; Gerisch *et al.*, 1975a).

During the preaggregation and early aggregation phases, a number of extracellular and cellular activities change dramatically. For example, shortly after starvation, the extracellular cAMP phosphodiesterase (ePD) activity decreases, and this drop in activity can be correlated with the secretion from the cells of a specific protein inhibitor (PDI) of this enzyme (Riedel *et al.*, 1973). These changes are followed closely by the appearance on the cell surface of several new activities, including contact sites A (Beug *et al.*, 1973), cAMP binding sites (Malchow and Gerisch, 1973; Green and Newell, 1975), and a membrane-bound cAMP phosphodiesterase (mPD) (Malchow *et al.*, 1972). It has also been reported that there is an appreciable increase in adenyl cyclase activity during this period (Klein, 1976; Roos *et al.*, 1977b).

¹ Abbreviations used: cAMP, adenosine 3',5'-cyclic monophosphate; ePD, extracellular cAMP phosphodiesterase; PDI, extracellular cAMP phosphodiesterase inhibitor; mPD, membrane-bound cAMP phosphodiesterase; DTT, dithiothreitol; Mes, 2-(*N*-morpholino)ethanesulfonic acid; TCA, trichloroacetic acid.

When amoebae, suspended in buffer, are treated with exogenous cAMP, in the form of either a single pulse or a series of pulses, the ePD activity of the suspension increases and PDI synthesis is inhibited (Klein, 1975; Tsang and Coukell, 1977; Klein and Darmon, 1977). (Since these two extracellular activities are regulated in a reciprocal fashion by exogenous cAMP, they will be referred to in this paper as the ePD-PDI system.) Under the same conditions, artificial periodic pulses of cAMP (but not cAMP added continuously) have been shown to accelerate the appearance on the cell surface of contact sites A, cAMP binding sites, and mPD (Gerisch *et al.*, 1975a) and to activate the adenylyl cyclase (Klein and Darmon, 1977; Roos *et al.*, 1977b). Based on the latter observation, Klein and Darmon (1977) suggested that the exogenous cAMP might regulate cell differentiation by stimulating adenylyl cyclase activity, thereby elevating the level of intracellular cAMP and enhancing endogenous signal emission.

To determine if the regulation of the ePD-PDI system and membrane differentiation by exogenous cAMP is mediated by a common mechanism involving the adenylyl cyclase, we have measured the different extracellular and cellular activities and the intracellular cAMP pools of cells treated with cAMP pulses and a continual flux of cAMP. We have also examined these parameters in two aggregation-deficient mutants which respond differently to exogenous cAMP. The results suggest that exogenous cAMP regulates the production of the extracellular and cellular activities by different mechanisms and that regulation of the ePD-PDI system by cAMP does not require the activation of the adenylyl cyclase.

MATERIALS AND METHODS

Chemicals. [8-³H]cAMP (21 Ci/mmole) and NCS tissue solubilizer were purchased from Amersham/Searle, Oakville, Ontario. cAMP, 5'-AMP, dithiothreitol (DDT), 2-

(*N*-morpholino)ethanesulfonic acid (Mes), and bovine serum albumin were from Sigma Chemical Co., St. Louis, Mo. Trichloroacetic acid (TCA) was from Fisher Scientific Co., Toronto, Ontario. 2-Ethoxyethanol (scintillation grade) was purchased from British Drug House, Toronto, Ontario.

Strains and culture conditions. Amoebae of *D. discoideum* strain AX2 (Watts and Ashworth, 1970) were cultured in HL-5 medium (Cocucci and Sussman, 1970) at 22°C on a Gallenkamp orbital shaker. Cells growing exponentially ($1-5 \times 10^6$ cells/ml) were harvested by centrifugation at 400g for 2 min. Wild-type strains HC6 and HC91 and the developmental mutants HC33, HC53, HC85, and HC112 have been described previously (Coukell, 1975). These strains were grown in association with *Klebsiella aerogenes* on SM agar plates (Sussman, 1966) and the amoebae were harvested in cold salt solution (SS) (Bonner, 1947) just as the bacterial lawns began to clear.

In all experiments, development was initiated by washing the cells twice in cold SS and once in 17 mM Sorenson's phosphate (SP) buffer, pH 6.0, and resuspending the cells in the same buffer at a concentration of 2×10^7 cells/ml. Zero time was defined as the time when the cells were resuspended in buffer.

Addition of cAMP to cell suspensions. One hundred milliliters of cell suspension in a 500-ml Erlenmeyer flask was shaken (≈ 120 rpm) in a New Brunswick gyrotary (Model G76) water bath shaker at 22°C. cAMP or phosphate buffer (for the control) was added dropwise to each flask by means of an electrically controlled peristaltic pump. The drops (33 μ l each) were added at 5-min intervals and the concentration of cAMP in the reservoir was adjusted to give the desired initial concentration in the cell suspension. When the nucleotide was to be added continuously, the flow rate was adjusted to deliver one drop every 30 sec and the delivery tubes were immersed in the cell suspension to ensure continuous addi-

tion. In these experiments, the concentration of cAMP in the reservoir was reduced so that the total amount of nucleotide added to the flask over a 5-min period was equivalent to a single 10 or 100 nM pulse. Even at the highest concentration of cAMP added, the level of ePD activity in the suspensions was more than sufficient to prevent an accumulation of the nucleotide. All activities, in these experiments, were corrected for cell density changes due to dilution.

cAMP binding, mPD, ePD, and PDI assays. All activities were determined on the same sample of cell suspension. At various times after the initiation of development, 5-ml aliquots of cell suspension were removed and centrifuged at 400g for 2 min. The cell pellet was placed on ice. The supernatant was further centrifuged at 10,000g for 10 min, dialyzed overnight at 4°C against two changes of 10 mM Tris-HCl buffer, pH 7.5, and stored at 4°C.

To measure the binding of [^3H]cAMP to intact cells, the cells were washed twice by centrifugation in cold 25 mM Mes buffer, pH 6.0, and assayed by the pellet assay method of Henderson (1975), except that 10 mM DTT was added to the rinsing buffer and the cGMP in the assay mixture was replaced by 20 mM DTT. All binding activities reported were averages of duplicate determinations after correction for nonspecific binding. Binding activity was expressed as picomoles of cAMP bound per 10^7 cells.

Cells not used in the binding assays were washed once more with 50 mM Tris-HCl buffer, pH 7.5, resuspended to 10^8 cells/ml in the same buffer, and stored at -20°C. When the samples were to be assayed for mPD activity, the cells were lysed by freezing and thawing (2 $\times$). The crude membrane preparation was then pelleted and washed twice in 50 mM Tris-HCl buffer, pH 7.5, by centrifugation (27,000g for 20 min) at 4°C. mPD activity was assayed as described by Malchow *et al.* (1972).

ePD activity was determined in the di-

alyzed supernatant fluids as described previously (Tsang and Coukell, 1977). One unit of ePD or mPD activity was defined as the amount of enzyme required to degrade 1 nmole of cAMP/min at 35°C.

To measure PDI activity, the dialyzed supernatant fractions were heated to 80°C for 30 min to inactivate the ePD and release the inhibitor. We found that heating the supernatant for only 10 min, as described previously (Riedel *et al.*, 1972), failed to inactivate all the ePD activity in samples containing very large amounts of enzyme. Inhibitor was assayed according to the method of Riedel *et al.* (1972) using ePD obtained from the inhibitor-negative strain HC35 (Coukell, 1975; Tsang and Coukell, unpublished), after 8 hr of starvation in SP buffer. Since this ePD preparation was rather unstable under these assay conditions (presumably due to the low level of protein secreted by HC35 during starvation), the inhibitor assay mixture was supplemented with bovine serum albumin to a final concentration of 500 $\mu\text{g}/\text{ml}$. One unit of PDI was defined as the amount of inhibitor required to inactivate one unit of ePD activity.

Protein concentration was determined according to the method of Lowry *et al.* (1951) with bovine serum albumin as a standard.

Determination of contact sites A. The time of appearance of contact sites A on the cell surface was determined as described by Darmon *et al.* (1975).

Extraction of intracellular cAMP. At each time point after the initiation of development, four 0.5-ml aliquots of cell suspension were removed at 90-sec intervals and collected on separate 25-mm Sartorius membrane filters (pore size, 0.8 μm). Each filter was washed with approximately 0.3 ml of SP buffer, and the four filters were immersed together in 2.0 ml of ice-cold 5% TCA to inactivate cell-associated phosphodiesterase activity. After 30 min on ice, the sample was heated to 95°C for 10 min, the filters were removed, and the TCA extract

was clarified by centrifugation at 27,000g for 15 min. One milliliter of the supernatant was then extracted 5× with 2 vol of water-saturated ether, and the residual ether was removed by boiling for 1 min. The extract was then frozen to -70°C , lyophilized, and redissolved in 0.5 ml of 50 mM acetate buffer, pH 4.0. The extracts were stored at -20°C until they were to be assayed. Recovery of cAMP by this procedure was about 80%. At all time points, greater than 90% of the protein kinase-reactive material in the TCA extracts was degraded after incubation with the ePD from strain HC35 for 4 hr at 25°C .

cAMP assay. cAMP was assayed according to Gilman (1970). The cAMP-dependent protein kinase and its inhibitor were isolated from fresh bovine heart muscle (Miyamoto *et al.*, 1969). An aliquot of the TCA extract was added to an assay mixture containing, in a total volume of 200 μl , 2.39 pmoles of [^3H]cAMP (21 Ci/mmol), 1 μg of protein kinase, 13.9 μg of protein kinase

inhibitor, and 50 mM acetate buffer, pH 4.0. After incubation at 0°C for 2 hr, 0.8 ml of cold 20 mM potassium phosphate buffer, pH 6.0, was added and the mixture was transferred quantitatively onto a 25-mm Sartorius membrane filter (pore size, 0.45 μm) which was then washed with 15 ml of the same buffer. Finally, the filters were dissolved in 1 ml of ethoxyethanol and 3 ml of scintillation fluid (toluene/PPO/POPOP) and counted. A blank containing only SP buffer was carried through the extraction procedure, assayed along with the cell extracts, and subtracted from each experimental value.

RESULTS

Effect of exogenous cAMP on membrane differentiation and regulation of the ePD-PDI system. Figure 1 shows the effect of pulses of cAMP, at an initial concentration of 100 nM, on the differentiation of amoebae of strain AX2 suspended in phosphate buffer. In this experiment, the time

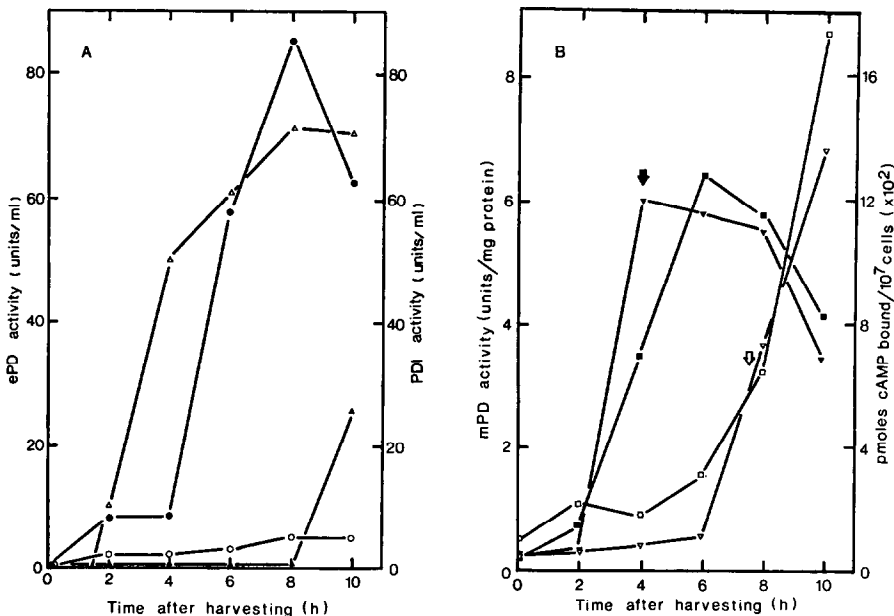


FIG. 1. Effect of 100 nM cAMP pulses on the ePD-PDI system (A) and membrane differentiation (B) of strain AX2. The cells were grown axenically, harvested, washed, shaken in phosphate buffer, treated with cAMP, and assayed as described in Materials and Methods. ePD activity (○, ●); PDI activity (△, ▲); mPD activity (▽, ▼); cAMP binding sites (□, ■); time of appearance of contact sites A (▢, ▣). Open symbols, untreated cells; closed symbols, cAMP-treated cells. The results are representative of three experiments.

of appearance of contact sites A, mPD, and cAMP binding sites on the cell surface was accelerated by 3–4 hr in the cAMP-treated suspension (Fig. 1B). These results confirm previous reports on the effect of cAMP pulses on membrane differentiation in *D. discoideum* (Gerisch *et al.*, 1975a; Roos *et al.*, 1977b).

In the same experiment, the ePD activity in the cAMP-treated suspension increased rapidly, reaching a level after 8 hr approximately 17-fold higher than the activity in the untreated suspension. This elevated level of ePD activity appeared to be due, at least in part, to the inhibition of PDI synthesis in the cAMP-pulsed cells (Fig. 1A). However, due to the presence of PDI in the control suspension, the low ePD activity measured in this suspension did not reflect accurately the amount of enzyme produced by the untreated cells; therefore it was not possible to determine from this experiment whether the cAMP pulses were also stimulating the rate of ePD synthesis. Since the PDI is a tight-binding inhibitor (Riedel *et al.*, 1972), Klein and Darmon (1975) have shown that it is possible to measure the total ePD (ePD_t) present in suspensions containing PDI by using the relationship

$ePD_t = ePD_d + (PDI_t - PDI_f)$, where ePD_d is the ePD activity detected, PDI_t is the total inhibitor and PDI_f is the free inhibitor in the suspension. The value $PDI_t - PDI_f$ represents the amount of ePD complexed with inhibitor. Using this approach, we repeated the experiment shown in Fig. 1 and calculated the total ePD present in an untreated suspension and in an identical suspension treated with 100 nM cAMP pulses. The results of one experiment are presented in Table 1. In the absence of cAMP pulses, total ePD production proceeded at a low rate for about 5 hr and then the rate increased sharply. This pattern of ePD production by untreated cells is very similar to that described by Klein and Darmon (1975). However, in the cAMP-pulsed suspension, the total ePD began to increase rapidly after only 1–2 hr, and after 5 hr of starvation there was approximately five times as much enzyme in the cAMP-treated suspension as in the control suspension. After this time, the rate of ePD production in the cAMP-treated suspension relative to the control suspension appeared to decrease slowly (see Table 1). These results indicate that the increased ePD activity detected in cell suspensions treated with

TABLE 1
EFFECT OF cAMP PULSES ON THE PRODUCTION OF EXTRACELLULAR cAMP PHOSPHODIESTERASE BY STRAIN AX2^a

Time (hr)	Control activities (units/ml)				cAMP-pulsed activities (units/ml)				Relative increase (B/A)
	ePD_d	PDI_t	PDI_f	ePD_t (A)	ePD_d	PDI_t	PDI_f	ePD_t (B)	
1	5.1	0	0	5.1	6.7	0	0	6.7	1.3
2	3.3	4.5	0	7.8	12.9	0	0	12.9	1.7
3	1.9	17.6	11.4	8.1	17.8	0	0	17.8	2.2
4	1.7	17.5	11.2	8.0	28.3	0	0	28.3	3.5
5	3.3	20.1	14.0	9.4	45.1	0	0	45.1	4.8
6	2.3	26.1	13.0	15.4	63.5	0	0	63.5	4.1
7	3.0	38.5	21.0	20.5	80.6	6.4	0	87.0	4.2
8	3.1	36.2	3.0	36.3	93.9	11.9	0	105.8	2.9

^a Amoebae were incubated and treated with cAMP as described in the legend to Fig. 1 and in Materials and Methods. The total ePD (ePD_t) in each supernatant sample was determined using the relationship $ePD_t = ePD_d + (PDI_t - PDI_f)$. ePD_d , enzyme activity detected in the supernatant; PDI_t , inhibitor activity measured after heating the supernatant at 80°C for 30 min; PDI_f , inhibitor activity in the unheated supernatant. These results are representative of three experiments.

cAMP pulses is due to *both* enhanced ePD synthesis and repressed PDI synthesis.

The effect of 1 nM cAMP pulses on cell differentiation is shown in Fig. 2. Although the concentration of the cAMP was reduced 100-fold, the small pulses were almost as effective as the large pulses at accelerating membrane differentiation (compare Figs. 1B and 2B). On the other hand, the 1 nM cAMP pulses seemed to be less effective at regulating the ePD-PDI system. When the cell suspensions were pulsed with 1 nM cAMP, the rate of PDI synthesis was only slightly reduced and the ePD activity of the pulsed suspensions increased slowly (Fig. 2A). Therefore, regulation of the ePD-PDI system by cAMP pulses would appear to be more concentration dependent than regulation of membrane differentiation.

Figure 3 shows the effect of the continuous addition of cAMP, at an average rate equivalent to the 100 nM pulses (see Materials and Methods), to amoebae of strain AX2 in suspension. Under these conditions, the addition of exogenous cAMP had no

effect on the rate of membrane differentiation (Fig. 3B; also see Gerisch *et al.*, 1975a), however, the ePD activity in the suspension increased dramatically and PDI synthesis was strongly inhibited (Fig. 3A). In addition to the effect of exogenous cAMP on ePD activity, total cellular PD activity (as determined on crude cell lysates) also increased rapidly after 1–2 hr of starvation (data not shown). Since the mPD activity was not elevated under these conditions (Fig. 3B), this activity would appear to represent nascent soluble enzyme. Furthermore, in this particular experiment, the ePD activity in the cAMP-treated suspension after 2 hr (before the appearance of an appreciable amount of inhibitor in the control) was four- to fivefold higher than the activity in the control suspension (Fig. 3A). Together, these observations suggest that the addition of a continuous flow of cAMP to cells in suspension inhibits PDI production and also induces the synthesis of ePD. The continuous addition of cAMP at a 10-fold lower concentration also failed to stim-

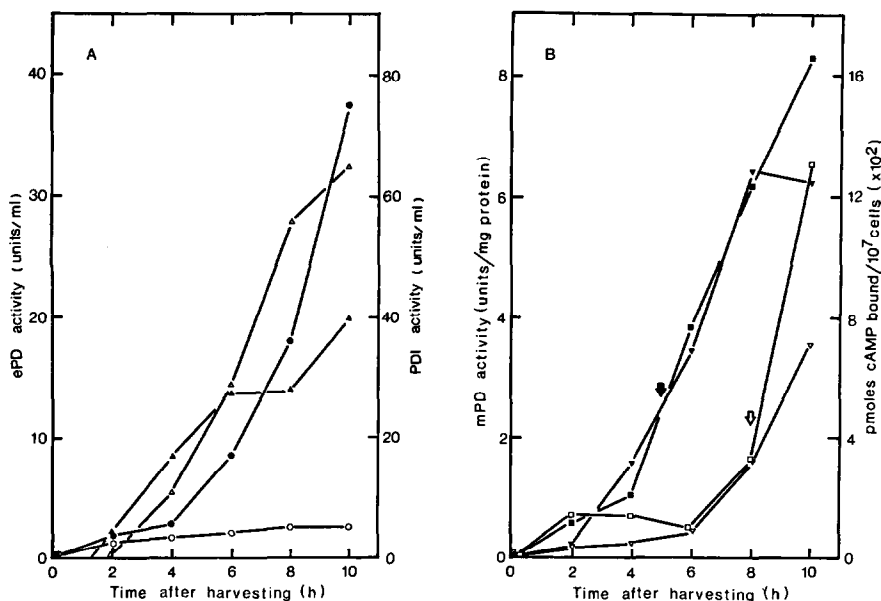


FIG. 2. Effect of 1 nM cAMP pulses on the ePD-PDI system (A) and membrane differentiation (B) of strain AX2. The cells were grown axenically, harvested, washed, shaken in phosphate buffer, treated with cAMP, and assayed as described in Materials and Methods. ePD activity (○, ●); PDI activity (△, ▲); mPD activity (▽, ▼); cAMP binding sites (□, ■); time of appearance of contact sites A (↓, ↓). Open symbols, untreated cells; closed symbols, cAMP-treated cells. The results are representative of two experiments.

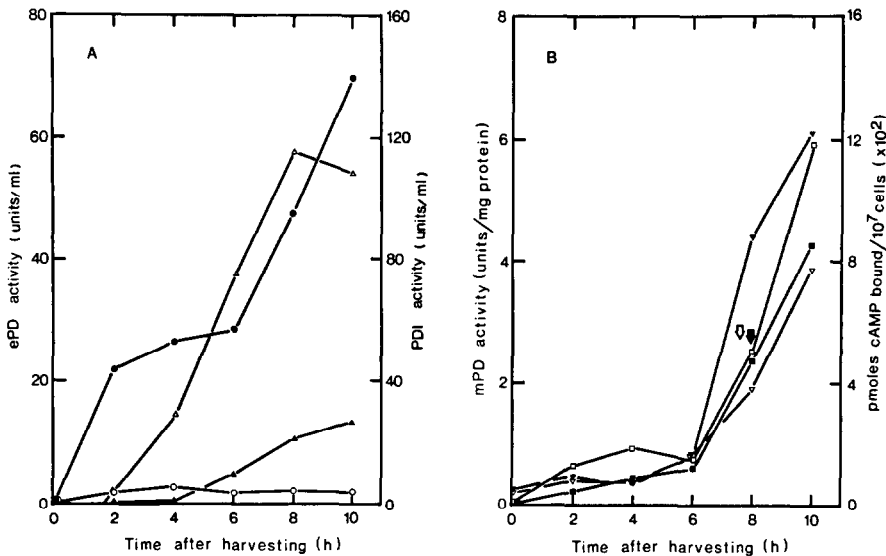


FIG. 3. Effect of the continuous addition of cAMP at an average rate equivalent to the 100 nM pulses on the ePD-PDI system (A) and membrane differentiation (B) of strain AX2. The cells were grown axenically, harvested, washed, shaken in phosphate buffer, treated with cAMP, and assayed as described in Materials and Methods. ePD activity (○, ●); PDI activity (△, ▲); mPD activity (▽, ▼); cAMP binding sites (□, ■); time of appearance of contact sites A (⏏, ⏏). Open symbols, untreated cells; closed symbols, cAMP-treated cells. The results are representative of three experiments.

ulate the rate of membrane differentiation, while the response of the ePD-PDI system under these conditions was comparable to the response produced by 1 nM pulses (data not shown).

Effect of exogenous cAMP on the level of intracellular cAMP. To determine if the exogenous cAMP was regulating the ePD-PDI system and/or membrane differentiation by activating the adenylyl cyclase, the concentration of the intracellular cAMP pools of untreated cells and cells treated with cAMP was monitored during early development.

When axenically grown amoebae of strain AX2 were shaken in phosphate buffer, the initial intracellular concentration of cAMP was 1–3 pmoles/ 10^7 cells, and this concentration remained relatively constant for about 2 hr before increasing sharply five- to sevenfold. After remaining at the elevated level for 1–2 hr, the cAMP concentration dropped to a level approximately two to three times the initial level (Fig. 4). Under the same conditions, similar

changes in the concentration of intracellular cAMP during development were observed with different wild-type strains grown on bacteria (see the results for strains HC91 and HC6 in Figs. 6 and 8, respectively).

When amoebae of strain AX2 were treated with 100 nM pulses of cAMP, the intracellular cAMP level remained relatively low for approximately 2 hr and then increased 20–25 fold, reaching a peak at 4 hr. Then, despite continued pulsing, the intracellular cAMP concentration dropped to a level comparable to that of the pool of untreated cells (Fig. 4). When the concentration of the artificial cAMP pulses was reduced to 10 or to 1 nM, the relative increase in the size of the intracellular cAMP pools was approximately the same (Table 2). However, when the cAMP was added in a continuous flux, at an average rate equivalent to the 100 nM pulses, not only was there no stimulation of the intracellular cAMP pool, but the normal increase in pool size observed in untreated cells was re-

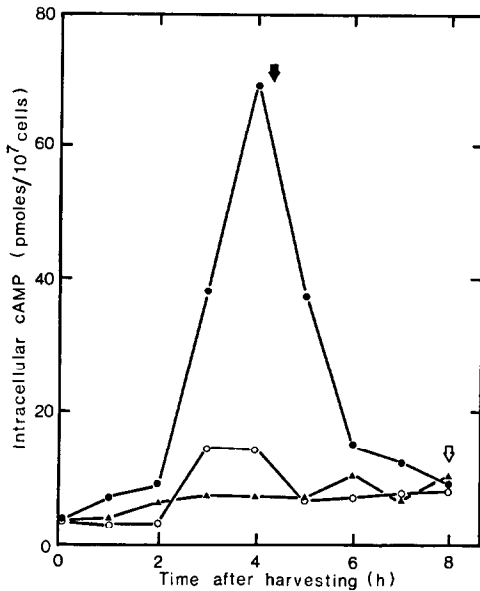


FIG. 4. Effect of exogenous cAMP on the concentration of intracellular cAMP in amoebae of strain AX2. The cells were grown axenically, harvested, washed, shaken in phosphate buffer, treated with cAMP, extracted, and assayed for cAMP as described in Materials and Methods. cAMP in untreated cells ($\circ$), cAMP-pulsed (100 nM) cells ($\bullet$), and cells treated with a continuous flux of cAMP at an average rate equivalent to the 100 nM pulses ($\blacktriangle$). The arrows indicate the time of appearance of contact sites A on untreated ($\Downarrow$) and on cAMP-pulsed cells ($\Downarrow$). The results are representative of three or more experiments under each set of conditions.

pressed (Fig. 4). Essentially the same result was obtained when the cAMP was added continuously at a 10-fold lower concentration, except that in some experiments, the level of the cAMP pool began to rise slowly after 6–8 hr of starvation (data not shown).

To estimate the amount of extracellular cAMP contaminating the intracellular cAMP pool determinations of cAMP-treated cells, the following control experiments were performed with strain AX2. At hourly intervals from 0 to 8 hr after starvation, 1.0-ml aliquots of cell suspension (2×10^7 cells/ml) were removed from the control flasks and treated with [^3H]cAMP to give a final concentration of 1–100 nM. Then the suspensions were agitated vigorously and the cells in 0.5 ml of the suspen-

sion were collected immediately on a filter, washed once with 0.3 ml of SP buffer, dissolved in 1.0 ml of ethoxyethanol, diluted with 3.0 ml of scintillation fluid, and counted. The results indicated that even at the highest concentration of cAMP used in our experiments (100 nM), contamination of the intracellular pools by extracellular cAMP was less than 0.3 pmoles/ 10^7 cells.

Effect of cAMP pulses on certain aggregation-deficient mutants. During the characterization of a group of aggregation-deficient mutants of *D. discoideum*, several mutants were found which failed to respond to exogenous cAMP in a variety of tests (Lo *et al.*, 1978). Strain HC112 failed to respond chemotactically to cAMP in the cellophane square test (Bonner *et al.*, 1966), and the developmental defect(s) in this strain could not be suppressed by artificial pulses of cAMP. The effect of cAMP pulses (100 nM) on membrane differentiation, the ePD-PDI system, and intracellular cAMP levels in strain HC112 is presented in Figs. 5 and 6. When amoebae of this strain were shaken in buffer, the cells produced relatively high levels of mPD and cAMP binding activities, however, the time of appearance and the maximum levels of these activities were the same in the presence and in the absence of cAMP pulses (Fig. 5B). Furthermore, neither the cAMP-treated nor the untreated cells formed detectable levels of contact sites A. The artificial cAMP pulses also had no effect on the intracellular cAMP pool of the mutant cells

TABLE 2
EFFECT OF ARTIFICIAL cAMP PULSES OF DIFFERENT CONCENTRATIONS ON THE INTRACELLULAR cAMP POOL OF AX2

Initial cAMP concentration ^a (nM)	Intracellular cAMP (pmoles/ 10^7 cells)		Relative increase
	Initial	Maximum	
100	3.0	69.4	23.1
10	3.5	68.8	19.6
1	2.5	46.1	18.4

^a Each experiment was performed with a different cell population.

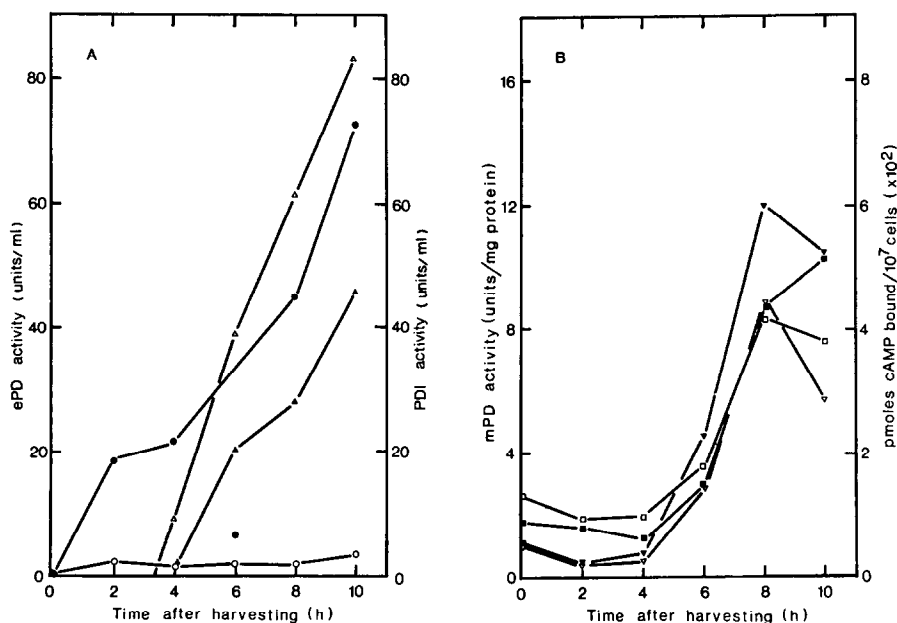


FIG. 5. Effect of 100 nM cAMP pulses on the ePD-PDI system (A) and membrane differentiation (B) of the aggregation-deficient mutant, strain HC112. The cells were grown on plates with bacteria, harvested, washed, shaken in phosphate buffer, treated with cAMP, and assayed as described in Materials and Methods. ePD activity (○, ●); PDI activity (△, ▲); mPD activity (▽, ▼); cAMP binding sites (□, ■). Open symbols, untreated cells; closed symbols, cAMP-pulsed cells. The results are representative of two experiments.

(Fig. 6). However, despite the absence of an effect on membrane differentiation and intracellular cAMP, the pulses of cAMP stimulated the level of ePD activity in the cAMP-treated suspension and partially inhibited PDI synthesis by these cells (Fig. 5A). Since the ePD activity in the cAMP-treated suspension, before the appearance of PDI, was approximately sevenfold higher than the activity in the control suspension, this elevated ePD activity would appear to be due to enhanced ePD synthesis as well as to a reduction in PDI production.

In addition to strain HC112, two other mutants, HC33 and HC85, failed to respond appreciably to exogenous cAMP (Lo *et al.*, 1978). When these strains were examined for the effect of artificial cAMP pulses on membrane differentiation, the level of intracellular cAMP and the ePD-PDI system, only the ePD-PDI system responded to the added nucleotide (data not shown). It should be noted that the developmental mutations in strains HC112 and HC33 have

been mapped at the same *agg* locus (*aggI*) (Coukell, 1975, 1977).

Another aggregation-deficient mutant, strain HC53, also failed to respond chemotactically to cAMP in the cellophane square test. However, unlike mutants HC112, HC33, and HC85, when amoebae of this strain were pulsed with cAMP (100 nM), the cells formed contact sites A within 6 hr, and when the cAMP-treated cells were placed on a solid substratum, they proceeded to form mature aggregates (Lo *et al.*, 1978). To determine if the developmental response of these cells to pulses of cAMP could be correlated with specific biochemical changes in the cells, the effect of cAMP pulses on the ePD-PDI system, membrane differentiation and intracellular cAMP levels was examined. The results are shown in Figs. 7 and 8. When amoebae of strain HC53, suspended in buffer, were treated with 100 nM pulses of cAMP, PDI synthesis was completely inhibited for 8 hr, and the ePD activity was elevated dramat-

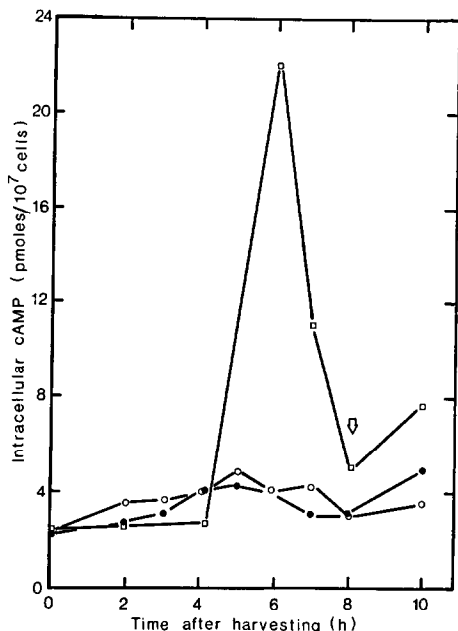


FIG. 6. Effect of 100 nM cAMP pulses on the concentration of intracellular cAMP in the aggregation-deficient mutant, strain HC112. The cells were grown on plates with bacteria, harvested, washed, shaken in phosphate buffer, treated with cAMP, extracted, and assayed for cAMP as described in Materials and Methods. For comparison, the intracellular cAMP pool of the wild-type parent of this mutant (strain HC91) is also shown. Untreated mutant cells (○); cAMP-pulsed mutant cells (●); untreated wild-type cells (□). The arrow (∥) indicates the time when contact sites A were first detected on strain HC91. The results are representative of two experiments.

ically (Fig. 7A). Therefore, the ePD and PDI activities in the mutant HC53 appeared to be regulated by exogenous cAMP in the same manner as they are regulated in wild-type cells. In addition, treatment of the mutant cells with cAMP pulses resulted in a sharp increase in mPD activity and cAMP binding sites after 4–6 hr of starvation, and in this experiment, contact sites A were detected on the cAMP-pulsed cells after ≈ 6 hr (Fig. 7B). In the absence of cAMP pulses, the concentration of the cAMP pool of the mutant cells remained low during the first 10 hr of development. When the cells were pulsed with cAMP (100 nM), the concentration of the cAMP pool was unaffected for ≈ 4 hr and then

increased abruptly, reaching a level comparable to the maximum level observed in the wild-type parent of this mutant, strain HC6. The concentration of the cAMP pool then decreased rapidly to a steady-state level, equivalent to that of the cAMP pools of the untreated mutant cells and the parental cells (Fig. 8).

DISCUSSION

The results obtained in this study suggest that, in *D. discoideum*, extracellular cAMP regulates the ePD–PDI system and the formation of certain membrane components by two functionally distinct mechanisms.

As reported previously (Gerisch *et al.*, 1975a), we have observed that the time of appearance of membrane components such as contact sites A, mPD, and cAMP binding sites was accelerated when cells suspended in buffer were treated with pulses of cAMP, but not by the continuous addition of cAMP (Figs. 1B and 3B). Furthermore, the degree of acceleration of membrane differentiation by cAMP pulses was approximately the same (3–4 hr), regardless of whether the concentration of the pulse was 100 or 1 nM (Figs. 1B and 2B). In addition, pulses of cAMP produced a dramatic increase in the concentration of the intracellular cAMP pools of the treated cells (Fig. 4), presumably by activating the plasma membrane-bound adenylyl cyclase (Klein and Darmon, 1977; Roos *et al.*, 1977b), and the magnitude of this increase was also relatively constant over a 1–100 nM range of cAMP concentrations. On the other hand, treatment of the cells with a continuous flux of cAMP not only failed to increase the level of the intracellular cAMP pool, but generally decreased the pool size (Fig. 4). These observations support the hypothesis (Gerisch *et al.*, 1975a; Roos *et al.*, 1977b) that certain events in early cell differentiation are regulated by a mechanism which detects a change in extracellular cAMP concentration with time, rather than simply an increase in the concentration of cAMP. The observation that small

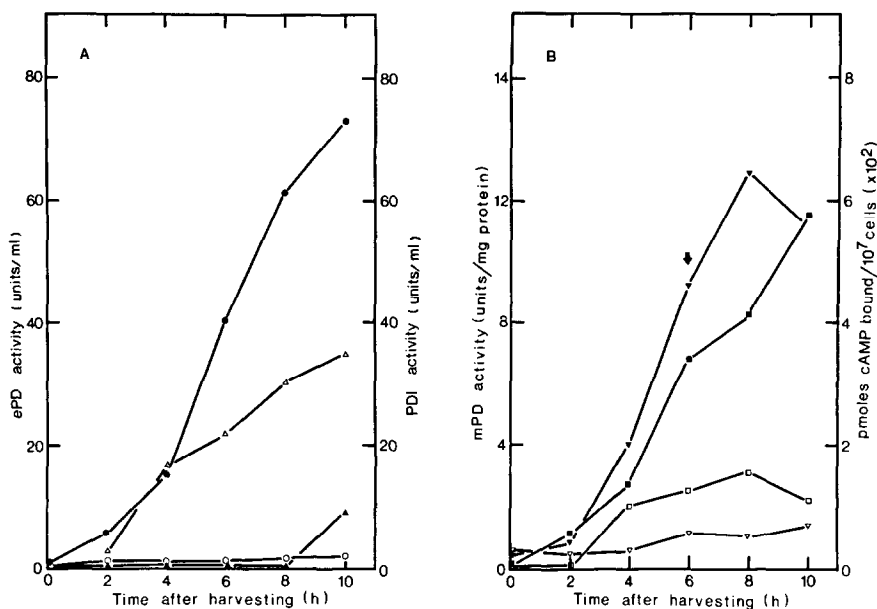


FIG. 7. Effect of 100 nM cAMP pulses on the ePD-PDI system (A) and membrane differentiation (B) of the aggregation-deficient mutant, strain HC53. The cells were grown on plates with bacteria, harvested, washed, shaken in phosphate buffer, treated with cAMP, and assayed as described in Materials and Methods. ePD activity (○, ●); PDI activity (△, ▲); mPD activity (▽, ▼); cAMP binding sites (□, ■). Open symbols, untreated cells; closed symbols, cAMP-pulsed cells. The arrow (↓) indicates the time when contact sites A were first detected on the cAMP-pulsed cells. The results are representative of two experiments.

pulses of cAMP (1 nM) are almost as effective as large pulses (100 nM) at increasing intracellular cAMP levels and at stimulating membrane differentiation also supports the suggestion that certain effects of exogenous cAMP pulses on cell differentiation are mediated through the activation of the adenylyl cyclase (Klein and Darmon, 1977; Roos *et al.*, 1977b). Further evidence for such a mechanism is provided by the results with the two aggregation-deficient mutants, HC53 and HC112. When amoebae of strain HC53 were pulsed with cAMP, the stimulation of membrane differentiation observed in this strain was accompanied by a fivefold increase in intracellular cAMP (Figs. 7B and 8), while under the same conditions, neither membrane differentiation nor the size of the cAMP pool of strain HC112 was stimulated when the cells were treated with cAMP (Figs. 5B and 6).

Unlike the effect of exogenous cAMP on membrane differentiation, the ePD-PDI system was regulated by *both* pulses of

cAMP and a continuous flux of cAMP (Figs. 1A and 3A). In addition, regulation of the ePD-PDI system by cAMP seemed to be concentration dependent, because when the concentration of the cAMP pulse was reduced 100-fold, there was an appreciable reduction in the rate of appearance of ePD activity in the cAMP-treated suspension and PDI synthesis was only slightly inhibited (Figs. 1A and 2A). The ePD-PDI system was also regulated by exogenous cAMP under experimental conditions (e.g., in the presence of a continuous flow of cAMP) which failed to produce an increase in intracellular cAMP levels (Figs. 3A and 4). Together, these results suggest that the ePD-PDI system is regulated by a mechanism which responds to elevated levels of extracellular cAMP, as well as to a change in the concentration of cAMP with time. Furthermore, exogenous cAMP appears to regulate this system without activating the adenylyl cyclase. The results of experiments with the developmental mutant

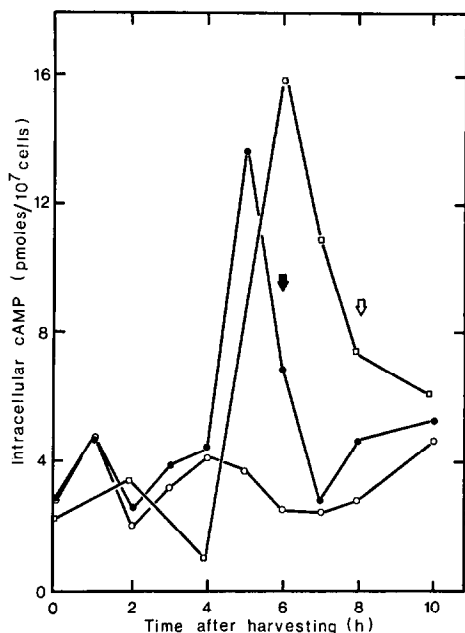


FIG. 8. Effect of 100 nM cAMP pulses on the concentration of intracellular cAMP in the aggregation-deficient mutant, strain HC53. The cells were grown on plates with bacteria, harvested, washed, shaken in phosphate buffer, treated with cAMP, extracted, and assayed for cAMP as described in Materials and Methods. For comparison, the intracellular cAMP pool of the wild-type parent of this mutant (strain HC6) is also shown. Untreated mutant cells (○); cAMP-pulsed mutant cells (●); untreated wild-type cells (□). The arrows indicate the time of appearance of contact sites A on the wild-type cells (⇓) and on the cAMP-pulsed mutant cells (↓). The results are representative of two experiments.

HC112 also indicated that membrane differentiation and the ePD-PDI system are regulated independently by extracellular cAMP. Membrane differentiation and the intracellular cAMP pool of the mutant cells was unaffected by 100 nM cAMP pulses, while the ePD-PDI system in this mutant was regulated normally, except that the inhibition of PDI synthesis was incomplete (Figs. 5 and 6). The properties of the mechanism described here for regulating the ePD-PDI system by exogenous cAMP are consistent with the proposed role of this system in early *D. discoideum* development (Tsang and Coukell, 1977), i.e., to maintain a low concentration of extracellular cAMP

in the vicinity of the preaggregative cells, thereby increasing the signal-to-noise ratio of the endogenous cAMP pulses.

Do measurements of the intracellular cAMP pools of amoebae treated with exogenous cAMP reasonably reflect the *in vivo* activity of the adenylyl cyclase? In shaken suspensions, the basal adenylyl cyclase activity of amoebae of strain AX2 has been reported to increase as much as 45-fold after approximately 3–4 hr of starvation (Klein, 1976; Klein and Darmon, 1977; Roos *et al.*, 1977b). Superimposed upon this increase in basal activity are rhythmic oscillations in adenylyl cyclase activity which also begin about this time. These oscillations occur with a periodicity of 5–8 min and have an amplitude 3–10 times that of the basal activity (Roos *et al.*, 1977a; Klein *et al.*, 1977). Together, these changes in adenylyl cyclase activity are probably responsible for the sharp increase in the level of the cAMP pool observed after 3–4 hr in amoebae of strain AX2 grown axenically (Fig. 4) and after 5–6 hr in amoebae of strains HC91 and HC6 grown on bacteria (Figs. 6 and 8). The subsequent drop in the concentration of the cAMP pool after 5–6 hr is probably due to enhanced secretion of the nucleotide by the cells, since extracellular cAMP levels are maximal at this time (Klein and Brachet, 1975). When amoebae, under the same conditions, are treated with artificial pulses of cAMP, both the basal activity and the periodic oscillations in adenylyl cyclase activity are stimulated after 3–4 hr of starvation (Roos *et al.*, 1977b; Gerisch and Hess, 1974; Roos *et al.*, 1977a). Presumably, this activation of the adenylyl cyclase by cAMP pulses is responsible for the dramatic increase in the cAMP pools of the cAMP-pulsed cells observed at this time (Fig. 4; Table 2). Again, the subsequent drop in intracellular cAMP is probably due to enhanced secretion. When cAMP was added to the cell suspensions in a continuous flow, the cAMP pools of the cells usually increased slowly two- to three-fold and then remained relatively constant

for 6–8 hr (see Fig. 4). Since it has been reported that the basal adenylyl cyclase activity is not activated by the continual addition of cAMP (Roos *et al.*, 1977b) and that cAMP applied in this form extinguishes the periodic oscillations in adenylyl cyclase activity (Gerisch and Hess, 1974; Roos *et al.*, 1977a), the modest increase in intracellular cAMP observed under these conditions is probably due to the appearance in the developing cells of nonactivated adenylyl cyclase (see Roos *et al.*, 1977b). Alternatively, the low level of intracellular cAMP detected in cells treated with a continual flux of cAMP might be an artifact due to the elevated cellular PD activity present under these conditions (Yeh and Coukell, unpublished). However, this interpretation seems unlikely since cellular PD levels are also elevated by treating the cells with cAMP pulses (Klein and Darmon, 1975; Lo and Coukell, unpublished), and under these conditions the concentration of the cAMP pool is very high (Fig. 4). In general, then, the changes observed in the levels of the cAMP pools of cells under different conditions do appear to reflect the changes occurring in adenylyl cyclase activity under the same conditions.

These results suggest that during the first few hours of starvation, the amoebae might form two distinct types of cAMP receptors: (1) a receptor which detects a change in the concentration of extracellular cAMP with time, and responds by inducing a number of changes in the plasma membrane (and possibly other events), probably by activating the adenylyl cyclase; and (2) a receptor which simply detects an increase in the level of extracellular cAMP and, without activating the adenylyl cyclase, increases the production of ePD and inhibits PDI synthesis. However, the results of experiments described in this study do not exclude the possibility that a single cAMP receptor could be involved in regulating both processes. During this study, we consistently observed a distinct peak of cAMP binding to the cell surface after approxi-

mately 2 hr of starvation (e.g., see Figs. 1B and 2B). Unlike the major cAMP binding sites, which reach a maximum activity after 8–10 hr in strain AX2, the appearance of these “early” binding sites is not stimulated by cAMP pulses and their appearance is not accompanied by an increase in mPD activity (Figs. 1B and 2B) (Lo *et al.*, 1978). The properties of these early sites together with their time of appearance during development suggest that these sites might be components of a receptor(s) system, which is formed shortly after starvation, to detect changes in the level of extracellular cAMP. We are presently examining these “early” binding sites for a possible role(s) in regulating early development.

F.K.C. was a recipient of a Province of Ontario Postgraduate Fellowship. This work was supported by a grant from the National Research Council of Canada.

REFERENCES

- BEUG, H., KATZ, F. E., and GERISCH, G. (1973). Dynamics of antigenic membrane sites relating to cell aggregation in *Dictyostelium discoideum*. *J. Cell Biol.* **56**, 647–658.
- BONNER, J. T. (1947). Evidence for the formation of cell aggregates by chemotaxis in development of the slime mold *Dictyostelium discoideum*. *J. Exp. Zool.* **106**, 1–26.
- BONNER, J. T. (1970). Induction of stalk cell differentiation by cyclic AMP in the cellular slime mold *Dictyostelium discoideum*. *Proc. Nat. Acad. Sci. USA.* **65**, 110–113.
- BONNER, J. T., KELSO, A. P., and GILLMOR, R. G. (1966). A new approach to the problem of aggregation in the cellular slime molds. *Biol. Bull.* **130**, 28–42.
- COCUCCI, S., and SUSSMAN, M. (1970). RNA in cytoplasmic and nuclear fractions of cellular slime mold amebas. *J. Cell Biol.* **45**, 399–407.
- COUKELL, M. B. (1975). Parasexual genetic analysis of aggregation-deficient mutants of *Dictyostelium discoideum*. *Mol. Gen. Genet.* **142**, 119–135.
- COUKELL, M. B. (1977). Evidence against mutational “hot-spots” at aggregation loci in *Dictyostelium discoideum*. *Mol. Gen. Genet.* **151**, 269–273.
- DARMON, M., BRACHET, P., and PEREIRA DA SILVA, L. H. (1975). Chemotactic signals induce cell differentiation in *Dictyostelium discoideum*. *Proc. Nat. Acad. Sci. USA* **72**, 3163–3166.
- GERISCH, G., and HESS, B. (1974). Cyclic-AMP-controlled oscillations in suspended *Dictyostelium* cells: Their relation to morphogenetic cell interactions.

- Proc. Nat. Acad. Sci. USA* **71**, 2118-2122.
- GERISCH, G., HUESGEN, A., and MALCHOW, D. (1975a). Genetic control of cell differentiation and aggregation in *Dictyostelium*: The role of cyclic-AMP pulses. In "Proceedings of the Tenth FEBS Meeting," pp. 257-267.
- GERISCH, G., HÜLSER, D., MALCHOW, D., and WICK, U. (1975b). Cell communication by periodic cyclic-AMP pulses. *Phil. Trans. Roy. Soc. London B* **272**, 181-192.
- GILMAN, A. C. (1970). A protein binding assay for adenosine 3',5'-cyclic monophosphate. *Proc. Nat. Acad. Sci. USA* **67**, 305-312.
- GREEN, A. A., and NEWELL, P. C. (1975). Evidence for the existence of two types of cAMP binding sites in aggregating cells of *Dictyostelium discoideum*. *Cell* **6**, 129-136.
- HENDERSON, E. J. (1975). The cyclic adenosine 3',5'-monophosphate receptor of *Dictyostelium discoideum*. *J. Biol. Chem.* **250**, 4730-4736.
- KLEIN, C. (1975). Induction of phosphodiesterase by cyclic adenosine 3',5'-monophosphate in differentiating *Dictyostelium discoideum* amoebae. *J. Biol. Chem.* **250**, 7125-7138.
- KLEIN, C. (1976). Adenylate cyclase activity in *Dictyostelium discoideum* amoebae and its changes during differentiation. *FEBS Lett.* **68**, 125-128.
- KLEIN, C., and BRACHET, P. (1975). Effects of progesterone and EDTA on cyclic AMP and phosphodiesterase in *Dictyostelium discoideum*. *Nature (London)* **254**, 432-434.
- KLEIN, C., and DARMON, M. (1975). The relationship of phosphodiesterase to the developmental cycle of *Dictyostelium discoideum*. *Biochem. Biophys. Res. Commun.* **67**, 440-447.
- KLEIN, C., and DARMON, M. (1977). Effects of cyclic AMP pulses on adenylate cyclase and the phosphodiesterase inhibitor of *D. discoideum*. *Nature (London)* **268**, 76-78.
- KLEIN, C., BRACHET, P., and DARMON, M. (1977). Periodic changes in adenylate cyclase and cAMP receptors in *Dictyostelium discoideum*. *FEBS Lett.* **76**, 145-147.
- KONIJN, T. M., BARKLEY, D. S., CHANG, Y. Y., and BONNER, J. T. (1968). Cyclic AMP: A naturally occurring acrasin in the cellular slime molds. *Amer. Natur.* **102**, 225-233.
- LO, E. K.-L., COUKELL, M. B., TSANG, A. S., and PICKERING, J. L. (1978). Physiological and biochemical characterization of aggregation-deficient mutants of *Dictyostelium discoideum*: Detection and response to exogenous cyclic AMP. *Canad. J. Microbiol.* **24**, 455-465.
- LOOMIS, W. F. (1975). "*Dictyostelium discoideum*: A Developmental System." Academic Press, New York/London.
- LOWRY, O. H., ROSEBROUGH, N. J., FARR, A. L., and RANDALL, R. J. (1951). Protein measurement with the Folin phenol reagent. *J. Biol. Chem.* **193**, 265-275.
- MALCHOW, D., and GERISCH, G. (1973). Cyclic AMP binding to living cells of *Dictyostelium discoideum* in the presence of excess cyclic GMP. *Biochem. Biophys. Res. Commun.* **55**, 200-204.
- MALCHOW, D., NÄGELE, B., SCHWARZ, H., and GERISCH, G. (1972). Membrane-bound cyclic AMP phosphodiesterase in chemotactically responding cells of *Dictyostelium discoideum*. *Eur. J. Biochem.* **28**, 136-142.
- MIYAMOTO, E., KUO, J. F., and GREENGARD, P. (1969). Cyclic nucleotide-dependent protein kinases. III. Purification and properties of adenosine 3',5'-monophosphate-dependent protein kinase from bovine brain. *J. Biol. Chem.* **244**, 6395-6402.
- RIEDEL, V., MALCHOW, D., GERISCH, G., and NÄGELE, B. (1972). Cyclic AMP phosphodiesterase interaction with its inhibitor of the slime mold *Dictyostelium discoideum*. *Biochem. Biophys. Res. Commun.* **46**, 279-287.
- RIEDEL, V., GERISCH, G., MÜLLER, E., and BEUG, H. (1973). Defective cyclic adenosine-3',5'-phosphate-phosphodiesterase regulation in morphogenetic mutants of *Dictyostelium discoideum*. *J. Mol. Biol.* **74**, 573-585.
- ROOS, W., SCHEIDEGGER, C., and GERISCH, G. (1977a). Adenylate cyclase activity oscillations as signals for cell aggregation in *Dictyostelium discoideum*. *Nature (London)* **266**, 259-260.
- ROOS, W., MALCHOW, D., and GERISCH, G. (1977b). Adenylate cyclase and the control of cell differentiation in *Dictyostelium discoideum*. *Cell Different.* **6**, 229-239.
- SUSSMAN, M. (1966). Biochemical and genetic methods in the study of cellular slime mold development. In "Methods in Cell Physiology" (D. Prescott, ed.), Vol. 2, pp. 397-410. Academic Press, New York.
- TSANG, A. S., and COUKELL, M. B. (1977). The regulation of cyclic AMP-phosphodiesterase and its specific inhibitor by cyclic AMP in *Dictyostelium*. *Cell Different.* **6**, 75-84.
- WATTS, D. J., and ASHWORTH, J. M. (1970). Growth of myxamoebae of the cellular slime mould *Dictyostelium discoideum* in axenic culture. *Biochem. J.* **119**, 171-174.